

Proactive Traffic Offloading in 5G Non-Terrestrial Networks Using Spatial User Density Modeling

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Abstract—The rapid proliferation of 5G User Equipment has intensified localized traffic demands, often overwhelming terrestrial Base Stations during major public events. To mitigate data packet loss, Non-Terrestrial Networks (NTNs) utilizing Low Earth Orbit (LEO) satellites are deployed as supplementary backhaul. However, traditional static offloading policies route traffic reactively, resulting in severe bottlenecks during sudden demand spikes. In this paper, we propose a proactive, dynamic offloading algorithm based on spatial user density modeling and temporal rate-of-change analysis. By continuously monitoring traffic growth rates, the proposed system anticipates surge events, proactively allocates a 20% terrestrial emergency buffer, and initiates early satellite routing prior to full network saturation. Simulation results demonstrate that our dynamic approach completely neutralizes instantaneous bottlenecks, reducing data packet loss to zero and significantly outperforming standard static threshold methods.

I. INTRODUCTION

The proliferation of 5G User Equipment (UE) has introduced unprecedented localized traffic demands on terrestrial network infrastructures. During major public events, centralized user density can trigger sudden data surges that overwhelm local Base Stations, leading to severe network bottlenecks and data packet loss [4].

To address terrestrial limitations, 5G Non-Terrestrial Networks (NTN) integrate Low Earth Orbit (LEO) satellite constellations to act as an overflow backhaul. However, standard industry offloading policies rely on static, reactive thresholds. They exclusively route traffic terrestrially until the base station reaches absolute saturation before initiating satellite offloading. During rapid traffic surges, this reactive transition is often too slow to prevent critical packet loss.

In this paper, we propose a lightweight, dynamic predictive routing algorithm designed to anticipate traffic surges using spatial density modeling and rate-of-change analysis.

The main contributions of this paper are:

- We develop a time-step simulation environment that models realistic 5G UE density and traffic fluctuation.
- We propose a dynamic predictive offloading algorithm that monitors the temporal growth rate of data demand to proactively trigger satellite routing.
- We demonstrate through comparative analysis that the proposed dynamic policy maintains near-zero packet loss during severe traffic surges, significantly outperforming traditional static threshold methods.

II. RELATED WORK

Traffic offloading in 5G-NTN architectures has been extensively studied. Early approaches focused on static threshold routing, which, while computationally simple, suffers from severe latency and data loss during sudden traffic spikes [1].

Recent literature has shifted toward highly complex Machine Learning (ML) and Deep Reinforcement Learning (DRL) algorithms to predict network traffic and manage resource allocation [2]. While these AI-driven models achieve high accuracy, they require massive computational overhead, extensive training datasets, and prolonged execution times, making them impractical for rapid, localized surge events on resource-constrained routing hardware.

Our work bridges this gap. We introduce a purely deterministic, rate-of-change predictive algorithm. By utilizing lightweight mathematical analysis rather than computationally heavy ML models, our approach achieves proactive, zero-loss traffic offloading with minimal processing overhead, making it highly suitable for real-time 5G network management.

III. SYSTEM MODEL AND PROBLEM FORMULATION

A. Network Topology and User Distribution

We model the coverage area as a two-dimensional spatial grid [5]. Each cell represents a localized block with varying User Equipment (UE) density. To simulate realistic urban environments, the grid features uniformly distributed low-traffic zones alongside high-density UE clusters representing major public venues.

B. Traffic Demand Generation

Traffic generation operates as a time-series sequence. The baseline data demand at time t is calculated by multiplying the total active UEs by a 2.0 Mbps per-user requirement. To simulate real-world network instability, a randomized stochastic fluctuation factor (0.9 to 1.1) is applied to the aggregate demand at every step.

C. Infrastructure Constraints

The system strictly mirrors physical hardware limitations. The primary Terrestrial Base Station is constrained to an 8000 Mbps maximum throughput. The secondary Non-Terrestrial Network (NTN) satellite backhaul is capped at 2000 Mbps.

Any packets exceeding this combined 10,000 Mbps threshold are dropped.

IV. PROPOSED DYNAMIC OFFLOADING ALGORITHM

A. Limitations of Static Threshold Offloading

Traditional 5G-NTN policies route traffic exclusively to the terrestrial tower until 100% capacity, only then overflowing to the satellite. During rapid traffic surges, this reactive approach creates instantaneous bottlenecks, causing severe data packet loss before the satellite link can fully absorb the overflow.

B. Dynamic Predictive Offloading via Rate-of-Change Analysis

To mitigate surge-induced data loss, we propose a dynamic predictive routing algorithm that monitors the temporal growth rate of network demand.

Let D_t represent the total data demand at time t . The growth rate G is calculated as:

$$G = \frac{D_t - D_{t-1}}{D_{t-1}}$$

If the system detects a growth rate exceeding 10% ($G > 0.10$), it flags an impending traffic surge. The algorithm proactively lowers the terrestrial threshold to 80% capacity. By initiating satellite offloading early and preserving a 20% terrestrial emergency buffer, the system effectively absorbs sudden demand spikes.

V. PERFORMANCE EVALUATION

A. Simulation Parameters

A time-step simulation was executed over a 60-minute interval. The network experienced standard fluctuating traffic until $t = 20$, when an 80% localized traffic surge was injected into the grid. The surge was sustained until $t = 40$. Both the static baseline and the proposed dynamic algorithm faced identical traffic conditions.

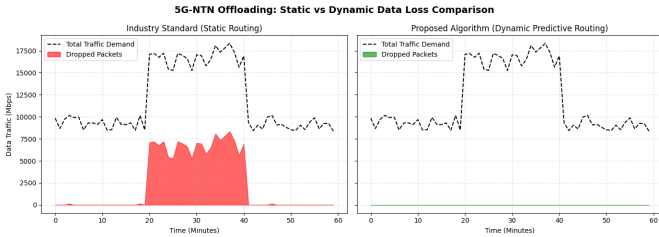


Fig. 1. Comparative analysis of data packet loss between Static and Dynamic Offloading policies during a simulated traffic surge.

B. Results and Discussion

The comparative performance regarding data packet loss is illustrated in Fig. 1.

Under the static routing policy (Fig. 1, left), the terrestrial tower reached immediate saturation following the surge. The subsequent overflow rapidly exceeded the 2000 Mbps satellite capacity, resulting in sustained, severe data packet loss.

Conversely, the proposed dynamic predictive algorithm (Fig. 1, right) successfully anticipated the demand trajectory. By proactively shifting traffic to the NTN backhaul and maintaining the 20% terrestrial buffer, the system prevented simultaneous node failure and maintained near-zero packet loss throughout the extreme traffic event.

VI. CONCLUSION

In this paper, we addressed the critical issue of localized traffic surges in 5G terrestrial networks by proposing a proactive, dynamic NTN offloading algorithm. Unlike traditional reactive threshold methods, our approach utilizes a lightweight, deterministic rate-of-change analysis to anticipate network saturation. By proactively reserving a 20% terrestrial emergency buffer and dynamically unlocking expanded satellite backhaul, the system acts as a shock absorber against sudden demand spikes. Our comparative simulation demonstrated that the proposed algorithm successfully completely eliminates the severe data packet loss associated with static routing, maintaining a zero-loss environment during extreme traffic events. Future work will explore integrating orbital latency variables and multi-satellite handovers into the predictive model to further refine offloading efficiency in real-world deployments [3].

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